

## **Lab Guide 5**

### **Managing Resources : Resources in Deployments, quota**

Q.1 Deploy Metrics Server

Q.2 Create a deployment with cpu and memory limits only with 200mb ram and 400m cpu value.

Q.3 Create a deployment with cpu and memory limits and requests also with any value.

Q.4 create a quota in your own namespace that pods=4, svc=2

Q.5 Create one Limit range with any value.

Ans: 1

kubectl top nodes

kubectl top pods

kubectl apply -f comp.yml

kubectl top pods

kubectl top nodes

```

root@master:~# kubectl top nodes
error: Metrics API not available
root@master:~# kubectl top pods
error: Metrics API not available
root@master:~# kubectl apply -f comp.yml
serviceaccount/metrics-server created
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader created
clusterrole.rbac.authorization.k8s.io/system:metrics-server created
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader created
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator created
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server created
service/metrics-server created
deployment.apps/metrics-server created
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io created
root@master:~# kubectl top pods
NAME      CPU(cores)   MEMORY(bytes)
pod       0m           7Mi
root@master:~# kubectl top nodes
NAME      CPU(cores)   CPU%    MEMORY(bytes)   MEMORY%
master    165m         8%      1138Mi          29%
worker    112m         5%      811Mi           21%
root@master:~# █

```

Ans: 2

vim dep.yaml

cat dep.yaml

kubectl create -f dep.yaml

kubectl describe deployment mayank

```

root@master:~# kubectl describe deployment mayank
Name: mayank
Namespace: jyothi
CreationTimestamp: Wed, 24 May 2023 08:26:42 +0000
Labels: app=mayank
Annotations: deployment.kubernetes.io/revision: 1
Selector: app=mayank
Replicas: 1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType: Recreate
MinReadySeconds: 0
Pod Template:
  Labels: app=mayank
  Containers:
    strategy:
      Image: httpd
      Port: <none>
      Host Port: <none>
      Requests:
        cpu: 400m
        memory: 200Mi
      Environment: <none>
      Mounts: <none>
      Volumes: <none>
  Conditions:
    Type          Status    Reason
    ----          -
    Available      True      MinimumReplicasAvailable
    Progressing    True      NewReplicaSetAvailable
  OldReplicaSets: <none>

```

```
root@master:~# cat dep.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  labels:
    app: mayank
  name: mayank
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mayank
  strategy:
    type: Recreate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: mayank
    spec:
      containers:
      - image: httpd
        name: strategy
        resources:
          requests:
            memory: 200Mi
            cpu: 400m
root@master:~#
```

Ans: 3

```
kubectl apply -f dep.yaml
```

```
kubectl describe deployment mayank
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  labels:
    app: mayank
  name: mayank
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mayank
  strategy:
    type: Recreate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: mayank
    spec:
      containers:
      - image: httpd
        name: strategy
        resources:
          requests:
            memory: 200Mi
            cpu: 400m
          limits:
            memory: 400Mi
            cpu: 600m
```

```
root@master:~# kubectl describe deployment mayank
Name: mayank
Namespace: jyothi
CreationTimestamp: Wed, 24 May 2023 08:26:42 +0000
Labels: app=mayank
Annotations: deployment.kubernetes.io/revision: 2
Selector: app=mayank
Replicas: 1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType: Recreate
MinReadySeconds: 0
Pod Template:
  Labels: app=mayank
  Containers:
    strategy:
      Image: httpd
      Port: <none>
      Host Port: <none>
      Limits:
        cpu: 600m
        memory: 400Mi
      Requests:
        cpu: 400m
        memory: 200Mi
```

Ans: 4

kubectl create deployment mayank --image httpd

kubectl create quota mayankquota --hard=pods=4,services=2

kubectl scale deployment mayank --replicas 5

kubectl get pods

```

root@master:~# kubectl create quota mayankquota --hard=pods=2,services=2
resourcequota/mayankquota created
root@master:~# kubectl get resourcequota
NAME          AGE    REQUEST          LIMIT
mayankquota    10s    pods: 1/2, services: 0/2
root@master:~# kubectl scale deployment mayank --replicas 5
deployment.apps/mayank scaled
root@master:~# kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
mayank-89b68f7f6-9z45t              0/1     ContainerCreating   0           2s
mayank-89b68f7f6-wk98z              1/1     Running             0          105s
root@master:~# kubectl describe deployment mayank
Name:                                mayank
Namespace:                          jyothis
CreationTimestamp:                   Wed, 24 May 2023 08:30:40 +0000
Labels:                              app=mayank
Annotations:                         deployment.kubernetes.io/revision: 1
Selector:                            app=mayank
Replicas:                            5 desired | 2 updated | 2 total | 2 available | 3 unavailable
StrategyType:                        RollingUpdate
MinReadySeconds:                     0
RollingUpdateStrategy:               25% max unavailable, 25% max surge
Pod Template:
  containers:
    image: jyothis/mayank
    name: mayank
    ports:
      containerPort: 80
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
  securityContext: {}
  serviceAccountName: default
  terminationGracePeriodSeconds: 30

```

Ans : 5

Cat limitrange.yml

apiVersion: "v1"

kind: "LimitRange"

metadata:

name: "resource-limits"

spec:

limits:

- type: "Container"

max:

cpu: "2"

memory: "1Gi"

min:

cpu: "100m"

memory: "4Mi"

default:

cpu: "300m"

memory: "200Mi"

kubectl create -f limitrange.yml

kubectl get limitrange

kubectl describe limitrange resource-limits

```
root@master:~# kubectl get limitrange
No resources found in default namespace.
root@master:~# cat limitrange.yml

apiVersion: "v1"
kind: "LimitRange"
metadata:
  name: "resource-limits"
spec:
  limits:
    - type: "Container"
      max:
        cpu: "2"
        memory: "1Gi"
      min:
        cpu: "100m"
        memory: "4Mi"
      default:
        cpu: "300m"
        memory: "200Mi"
root@master:~#
root@master:~#
root@master:~# kubectl create -f limitrange.yml
limitrange/resource-limits created
root@master:~# kubectl get limitrange
NAME                               CREATED AT
resource-limits 2023-10-04T05:26:56Z
root@master:~# kubectl describe limitrange
Name:          resource-limits
Namespace:     default
Type           Resource  Min    Max    Default Request  Default Limit  Max Limit/Request Ratio
-----
Container      memory    4Mi    1Gi    200Mi             200Mi           -
Container      cpu       100m    2      300m              300m            -
root@master:~#
```